

Difference-in-Differences with Multiple Time Periods*

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Abstract

In this article, we consider identification, estimation, and inference procedures for treatment effect parameters using Difference-in-Differences (DiD) with (i) multiple time periods, (ii) variation in treatment timing, and (iii) when the “parallel trends assumption” holds potentially only after conditioning on observed covariates. We show that a family of causal effect parameters are identified in staggered DiD setups, even if differences in observed characteristics create non-parallel outcome dynamics between groups. Our identification results allow one to use outcome regression, inverse probability weighting, or doubly-robust estimands. We also propose different aggregation schemes that can be used to highlight treatment effect heterogeneity across different dimensions as well as to summarize the overall effect of participating in the treatment. We establish the asymptotic properties of the proposed estimators and prove the validity of a computationally convenient bootstrap procedure to conduct asymptotically valid simultaneous (instead of pointwise) inference. Finally, we illustrate the relevance of our proposed tools by analyzing the effect of the minimum wage on teen employment from 2001–2007. Open-source software is available for implementing the proposed methods.

JEL: C14, C21, C23, J23, J38.

Keywords: Difference-in-Differences, Dynamic Treatment Effects, Doubly Robust, Event Study, Variation in Treatment Timing, Treatment Effect Heterogeneity, Semi-Parametric.

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1 Introduction

Difference-in-Differences (DiD) has become one of the most popular research designs used to evaluate causal effects of policy interventions. In its canonical format, there are two time periods and two groups: in the first period no one is treated, and in the second period some units are treated (the treated group), and some units are not (the comparison group). If, in the absence of treatment, the average outcomes for treated and comparison groups would have followed parallel paths over time (which is the so-called parallel trends assumption), one can estimate the average treatment effect for the treated subpopulation (ATT) by comparing the average change in outcomes experienced by the treated group to the average change in outcomes experienced by the comparison group. Methodological extensions of DiD methods often focus on this standard two periods, two groups setup; see, e.g., [Heckman et al. \(1997, 1998\)](#), [Abadie \(2005\)](#), [Athey and Imbens \(2006\)](#), [Qin and Zhang \(2008\)](#), [Bonhomme and Sauder \(2011\)](#), [de Chaisemartin and D’Haultfœuille \(2017\)](#), [Botosaru and Gutierrez \(2018\)](#), [Callaway et al. \(2018\)](#), and [Sant’Anna and Zhao \(2020\)](#).¹

Many DiD empirical applications, however, deviate from the canonical DiD setup and have more than two time periods and variation in treatment timing. In this article, we provide a unified framework for average treatment effects in DiD setups with multiple time periods, variation in treatment timing, and when the parallel trends assumption holds potentially only after conditioning on observed covariates. We concentrate our attention on DiD with staggered adoption, i.e., to DiD setups such that once units are treated, they remain treated in the following periods.

The core of our proposal relies on separating the DiD analysis into three separate steps: (i) identification of policy-relevant disaggregated causal parameters; (ii) aggregation of these parameters to form summary measures of the causal effects; and (iii) estimation and inference about these different target parameters. Our approach allows for estimation and inference on interpretable causal parameters allowing for arbitrary treatment effect heterogeneity and dynamic effects, thereby completely avoiding the issues of interpreting results of standard two-way fixed effects (TWFE) regressions as causal effects in DiD setups as pointed out by [Borusyak and Jaravel \(2017\)](#), [de Chaisemartin and D’Haultfœuille \(2020\)](#), [Goodman-Bacon \(2019\)](#), [Sun and Abraham \(2020\)](#), and [Athey and Imbens \(2018\)](#). In addition, it adds transparency and objectivity to the analysis ([Rubin \(2007, 2008\)](#)), and allows researchers to exploit a variety of estimation methods to answer different questions of interest.

The identification step of the analysis provides a blueprint for the other steps. In this paper, we pay particular attention to the disaggregated causal parameter that we call the *group-time average treatment effect*, i.e., the average treatment effect for group g at time t , where a “group” is defined by the time period when units are first treated. In the canonical DiD setup with two periods and two groups, these parameters reduce to the ATT which is typically the parameter of interest in that setup. An attractive feature of the group-time average treatment effect parameters is that they do not directly restrict heterogeneity with respect to observed covariates, the period in which units are first treated, or the evolution of treatment effects over time. As a consequence, these easy-to-interpret causal parameters can be directly used for learning about treatment effect heterogeneity, and/or to construct many other more aggregated causal

¹See Section 6 of [Athey and Imbens \(2006\)](#) and Theorem S1 in [de Chaisemartin and D’Haultfœuille \(2017\)](#) for notable exceptions that cover multiple periods and multiple groups.

Theorem 1. *Let Assumptions 1, 2, 3 and 6 hold.*

(i) *If Assumption 4 holds, then, for all g and t such that $g \in \mathcal{G}_\delta$, $t \in \{2, \dots, \mathcal{T} - \delta\}$ and $t \geq g - \delta$,*

$$ATT(g, t) = ATT_{ipw}^{nev}(g, t; \delta) = ATT_{or}^{nev}(g, t; \delta) = ATT_{dr}^{nev}(g, t; \delta).$$

(ii) *If Assumption 5 holds, then, for all g and t such that $g \in \mathcal{G}_\delta$, $t \in \{2, \dots, \mathcal{T} - \delta\}$ and $g - \delta \leq t < \bar{g} - \delta$,*

$$ATT(g, t) = ATT_{ipw}^{ny}(g, t; \delta) = ATT_{or}^{ny}(g, t; \delta) = ATT_{dr}^{ny}(g, t; \delta).$$

Theorem 1 is the first main result of this paper. It provides powerful identification results that extend the DiD identification results based on the outcome regression approach of Heckman et al. (1997, 1998), the IPW approach of Abadie (2005), and the DR approach of Sant’Anna and Zhao (2020) to the multiple-periods, multiple groups setup. In other words, Theorem 1 says that, from an identification point of view, one can recover the $ATT(g, t)$ ’s by exploiting different parts of the data generating process: the OR approach only relies on modeling the conditional expectation of the outcome evolution for the comparison groups, the IPW approach relies on modeling the conditional probability of being in group g , whereas the DR approach exploits both OR and IPW components.

In order to extend the results of Heckman et al. (1997, 1998), Abadie (2005), and Sant’Anna and Zhao (2020) to the multiple groups, multiple periods framework, we have to address two different challenges: one associated with an appropriate reference time period and one associated with an appropriate comparison group. Theorem 1 highlights how a solution to these challenges is directly connected to the limited anticipation and the conditional parallel trends assumptions. More specifically, Theorem 1 says that we can use the time period $t = g - \delta - 1$ as an appropriate reference time period under Assumption 3 and either Assumption 4 or 5. This is the most recent time period when untreated potential outcomes are observed for units in group g . Interestingly, the more treatment anticipation is allowed (i.e., the higher δ is), the further back in time one needs to go.⁹ Theorem 1 also suggests that the choice of comparison group is directly tied to the conditional parallel trends assumption one makes: under Assumption 4, one can use “never treated” units as a fixed comparison group for all “eventually treated” units; whereas, under Assumption 5, one can use the “not-yet-treated by time $t + \delta$ ” units as a valid comparison group for those who are first treated at time g . In this latter case, Theorem 1 also highlights that when all units eventually gets treated ($\bar{g} < \infty$), one is only able to identify the $ATT(g, t)$ ’s for time periods before the last treated group “effectively” starts their treatment, i.e., $t < \bar{g} - \delta$. In this case, one can not identify the $ATT(g, t)$ for the last treated cohort, too.

Finally, we note that when pre-treatment covariates play no role in identification (i.e., Assumptions 3, 4, and 5 hold unconditionally on X), (2.2)-(2.4) collapse to

$$ATT_{unc}^{nev}(g, t; \delta) = E[Y_t - Y_{g-\delta-1} | G_g = 1] - E[Y_t - Y_{g-\delta-1} | C = 1], \quad (2.8)$$

and (2.5)-(2.7) collapse to

$$ATT_{unc}^{ny}(g, t; \delta) = E[Y_t - Y_{g-\delta-1} | G_g = 1] - E[Y_t - Y_{g-\delta-1} | D_{t+\delta} = 0]. \quad (2.9)$$

⁹As mentioned in Remark 1, as one allows δ to increase, Assumptions 4 and 5 becomes more restrictive.

These expressions for $ATT(g, t)$ clearly resemble the one for ATT in the canonical two-periods and two-groups case. As in that case, the average effect of participating in the treatment for units in group g is identified by taking the path of outcomes (i.e., the change in outcomes between the most recent period before they were affected by the treatment and the current period) actually experienced by that group and adjusting it by the path of outcomes experienced by a comparison group. Under the parallel trends assumption, this latter path is the path of outcomes that units in group g would have experienced if they had not participated in the treatment.

Remark 3. From (2.8) one can see that when Assumptions 3 and 4 hold unconditionally and there is no-anticipation, the $ATT(g, t)$ parameter can be obtained by first subsetting the data to only contain observations at time t and $g - 1$, from units with either $G_g = 1$ or $C = 1$, and then, using only the observations of this subset, running the (population) linear regression

$$Y = \alpha_1^{g,t} + \alpha_2^{g,t} \cdot G_g + \alpha_3^{g,t} \cdot 1\{T = t\} + \beta^{g,t} \cdot (G_g \times 1\{T = t\}) + \epsilon^{g,t}. \quad (2.10)$$

It is then easy to verify that $\beta^{g,t} = ATT(g, t)$. Note that one would need to consider different partitions of the data to characterize different $ATT(g, t)$ in terms of regression parameters. Alternatively, one could use the interacted two-way fixed effects regression proposed by Sun and Abraham (2020).

Remark 4. When covariates are available, the $\tilde{\beta}^{g,t}$ coefficient of the population linear regression

$$Y = \tilde{\alpha}_1^{g,t} + \tilde{\alpha}_2^{g,t} \cdot G_g + \tilde{\alpha}_3^{g,t} \cdot 1\{T = t\} + \tilde{\beta}^{g,t} \cdot (G_g \times 1\{T = t\}) + \tilde{\gamma} \cdot X + \tilde{\epsilon}^{g,t}$$

that uses the same subset of data as in Remark 3 is, in general, not equal to $ATT(g, t)$ unless one is willing to assume (i) homogeneous (in X) treatment effects, i.e., $\mathbb{E}[Y_t(g) - Y_t(0)|G_g = 1, X] = \mathbb{E}[Y_t(g) - Y_t(0)|G_g = 1]$ a.s., and (ii) rule-out covariate-specific trends, i.e., for $\mathbb{E}[Y_t - Y_{t-1}|X, G] = \mathbb{E}[Y_t - Y_{t-1}|G]$ a.s. for all groups and time periods; see, e.g., Słoczyński (2018) for a related discussion. The characterizations of $ATT(g, t)$ discussed in Theorem 1 do not rely on these restrictive conditions.

Remark 5. Although the IPW, OR, and DR based estimands presented in Theorem 1 are identical from an identification standpoint, this is not the case when one wants to estimate and make inference about the $ATT(g, t)$. As we discuss in Section 4, DiD estimators based on the DR estimands (2.4) and (2.7) usually enjoy additional robustness against model-misspecifications when compared to the IPW and OR estimands.

Remark 6. Theorem 1 suggests that we can identify $ATT(g, t)$ only for groups in $\mathcal{G}_\delta \subseteq \mathcal{G}$ which can involve dropping some “early treated” groups due to anticipation effects. When $\delta = 0$, i.e. when there is no anticipation, $\mathcal{G}_\delta = \mathcal{G}$. Theorem 1 also suggests that we can identify $ATT(g, t)$ only until $t = \mathcal{T} - \delta$ because of potential treatment anticipation behavior. In applications where some units are known to never participate in the treatment (including periods after time period $\mathcal{T}$), however, we can identify $ATT(g, t)$ up to $t = \mathcal{T}$ by using these units as a valid comparison group for all time periods $t = \mathcal{T} - \delta + 1, \dots, \mathcal{T}$, provided that an appropriate parallel trends assumption is satisfied.

Remark 7. From Theorem 1 it is clear that pre-treatment covariates play a prominent role in our analysis. Importantly, Assumptions 4 and 5 suggest that researchers should include pre-treatment covariates

that are potentially associated with the outcome evolution of $Y(0)$ during post-treatment periods. We explicitly rule out incorporating post-treatment covariates as they can potentially be affected by the treatment; see, e.g., [Wooldridge \(2005b\)](#), for a related discussion under the unconfoundedness setup.

3 Summarizing Group-Time Average Treatment Effects

The previous section shows that we can identify the $ATT(g, t)$'s by restricting treatment anticipation behavior and imposing a conditional parallel trends assumption. In many applications, the $ATT(g, t)$'s can be the ultimate causal parameters of interest. They can be used to highlight treatment effect heterogeneity across different groups g , at different points in time t , and across different lengths of treatment exposure, $e = t - g$. In other situations, however, researchers may want to combine these different $ATT(g, t)$'s to form more aggregated causal parameters. For instance, if the number of groups and time periods is relatively large, it may be challenging to interpret many group-time average treatment effects.

In this section, we consider different aggregation schemes for the $ATT(g, t)$'s that allow researchers to form a variety of summary measures of the causal effects of a given policy. Our aggregation schemes are of the form

$$\theta = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\tau} w(g, t) \cdot ATT(g, t), \quad (3.1)$$

where $w(g, t)$ are carefully-chosen (known or estimable) weighting functions specified by the researcher such that θ can be used to address a well-posed empirical/policy question. Different choices of $w(g, t)$ allows researchers to highlight different types of treatment effect heterogeneity. We pay particular attention to aggregations that result in a single overall treatment effect summary parameter as well as to aggregations related to understanding dynamic effects as is commonly done in event-study analysis. Of course, many other aggregated parameters of the type (3.1) can be easily constructed following our framework. We illustrate this point by also summarizing heterogeneity with respect to group or by calendar time.

Before proceeding with the discussion on how to construct these different aggregated parameters, it is worth revisiting the two most popular treatment effect summary measures used by practitioners in DiD setups. These are based on the “static” and “dynamic” two-way fixed effects (TWFE) linear regression specifications

$$Y_{i,t} = \alpha_t + \alpha_g + \beta D_{i,t} + \epsilon_{i,t}, \quad (3.2)$$

$$Y_{i,t} = \alpha_t + \alpha_g + \sum_{e=-K}^{-2} \delta_e^{anticip} \cdot D_{i,t}^e + \sum_{e=0}^L \beta_e \cdot D_{i,t}^e + v_{i,t}, \quad (3.3)$$

respectively, where α_t is a time fixed effect, α_g is a group fixed effect, $\epsilon_{i,t}$ and $v_{i,t}$ are error terms, $D_{i,t}^e = 1\{t - G_i = e\}$ is an indicator for unit i being e periods away from initial treatment at time t , and K and L are positive constants. The parameter of interest in the static TWFE specification is β , which, in applications, is typically interpreted as an overall effect of participating in the treatment across groups and time periods. In the dynamic TWFE specification, practitioners usually focus on the β_e , $e \geq 0$, and these parameters are typically interpreted as measuring the effect of participating in the

treatment at different lengths of exposure to the treatment.

Despite the popularity of these specifications, recent research has shown that one must be very careful in attaching a causal interpretation to these aggregated parameters. For instance, [Borusyak and Jaravel \(2017\)](#), [Goodman-Bacon \(2019\)](#), [de Chaisemartin and D’Haultfœuille \(2020\)](#), and [Athey and Imbens \(2018\)](#) have shown that, in general, β recovers a weighted average of some underlying treatment effect parameters but some of the weights on these parameters can be negative. This can potentially lead to particularly problematic cases such as the effect of the treatment being positive for all units, but the TWFE estimation resulting in estimates of β that are negative. Even in cases where the weights are not negative, the weights on underlying treatment effect parameters are entirely driven by the TWFE estimation strategy and are sensitive to the size of each group, the timing of treatment, and the total number of time periods (see Theorem 1 in [Goodman-Bacon \(2019\)](#)). The results in this section can be used in exactly the same setup to identify a single interpretable average treatment effect parameter and, thus, provide a way to circumvent the issues with the more common approach.

As discussed by [Goodman-Bacon \(2019\)](#), the “negative weight problem” associated with β arises when treatment effects evolve over time. Thus, one may wonder if such problems would still be present when considering more general, dynamic specifications such as (3.3). [Sun and Abraham \(2020\)](#) shows that this is still the case as the β'_e s associated with (3.3) do not recover easy-to-interpret causal parameters and still generally suffer from the same sorts of “negative weighting problems.” In contrast to this, we provide a simple way to directly aggregate our group-time average treatment effects into average treatment effects across different lengths of exposure to the treatment.

3.1 Aggregations to Highlight Treatment Effect Heterogeneity

Next, we discuss several partial aggregations of the group-time average treatment effects in order to summarize different dimensions of treatment effect heterogeneity. Although there are additional possibilities, we focus our discussion below on how to answer three particular questions: (a) How does the effect of participating in the treatment vary with length of exposure to the treatment? (b) Do groups that are treated earlier have, on average, higher/lower average treatment effects than groups that are treated later? (c) What is the cumulative average treatment effect of the policy across all groups until some particular point in time? Throughout this section, to avoid notation clutter, we assume that units do not anticipate treatment, i.e., we consider the case where Assumption 3 holds with $\delta = 0$. We also assume that a “never treated” group is available.

3.1.1 How do average treatment effects vary with length of exposure to the treatment?

One of the most popular questions that arises in DiD setups with multiple time periods concerns treatment effect dynamics: How does the effect of participating in the treatment vary with length of exposure to the treatment? For instance, do average treatment effects increase/decrease with elapsed treatment time? Indeed, answering this type of question is often the main motivation for using the event study regression in (3.3), though, as we mentioned above, that sort of regression may not be suitable for such a task. In this section, we propose an aggregation scheme that is suitable to highlight treatment effect heterogeneity with respect to length of exposure to the treatment that does not suffer from the drawbacks associated